

Functions and Function Notation (Part II) Homework

① If $\frac{0}{x} = 3^{x+4}$, find $\frac{0}{-2}$, $\frac{0}{-1}$, and $\frac{0}{-4}$.

② If $\boxed{b}y = b - y$, find $\boxed{5}6$, $\boxed{7}4$, and $\boxed{-3}-8$.

③ If $f(x) = 3x$, find $f(0)$.

④ If $g(x) = 1 - 2^{2x}$, find $g(1)$.

⑤ If $h(x) = x^2 - 2x + 5$, find $h(-2)$.

For problem #'s 6-7, find the following:

A. $(f+g)(x)$ B. $(f-g)(x)$

C. $(f \cdot g)(x)$ D. $(\frac{f}{g})(x)$

E. $(f+g)(-3)$ F. $(f-g)(5)$

G. $(f \cdot g)(2)$ H. $(\frac{f}{g})(1)$

⑥ $f(x) = 2x - 1$ and $g(x) = x^2$

⑦ $f(x) = x + x^3$ and $g(x) = \frac{1}{x}$.

⑧ Find the following values using the functions in problem #6.

A. $(f \circ g)(x)$ B. $(g \circ f)(x)$

C. $(f \circ g)(2)$ D. $(g \circ f)(-3)$

⑨ Find the following values using the functions in problem #7.

A. $(f \circ g)(x)$

B. $(g \circ f)(x)$

C. $(f \circ g)(4)$

D. $(g \circ f)(-1)$

Given the functions below, find the values.

⑩ $f(x) = \begin{cases} x+4, & 5 < x < 10 \\ \frac{x}{3}, & 10 \leq x \leq 12 \end{cases}$

i) $f(9)$

ii) $f(11)$

iii) $f(12)$

iv) $f(10)$

v) $f(0)$

vi) $f(5)$

⑪ $f(x) = \begin{cases} 3^{x-1}, & -4 \leq x \leq 2 \\ -2x, & 5 < x \leq 8 \end{cases}$

i) $f(6)$

ii) $f(0)$

iii) $f(4)$

iv) $f(8)$

v) $f(2)$

vi) $f(5)$

⑫ $f(x) = \begin{cases} \frac{3}{x^2}, & -10 < x \leq -1 \\ 10 - 4x, & -1 < x < 7 \end{cases}$

i) $f(0)$

ii) $f(-2)$

iii) $f(1)$

iv) $f(-1)$

v) $f(8)$

vi) $f(7)$

Answers

① $\frac{0}{-2} = \boxed{0}$, $\frac{0}{-1} = \boxed{0}$, $\frac{0}{-4} = \boxed{0}$

② $\frac{5}{\triangle} 6 = \boxed{-1}$, $\frac{7}{\triangle} 4 = \boxed{3}$, $\frac{-3}{\triangle} -8 = \boxed{5}$.

③ $\boxed{0}$

④ $\boxed{-3}$

⑤ $\boxed{13}$

⑥ A. $\boxed{2x-1+x^2}$ The order of these terms does not matter.

B. $\boxed{2x-1-x^2}$

C. $\boxed{(2x-1)(x^2)}$ or $\boxed{2x^3-x^2}$

D. $\boxed{\frac{2x-1}{x^2}}$

E. $\boxed{2}$

F. $\boxed{-16}$

G. $\boxed{12}$

H. $\boxed{1}$

⑦ A. $\boxed{x+x^3+\frac{1}{x}}$

B. $\boxed{x+x^3-\frac{1}{x}}$

C. $\boxed{(x+x^3)(\frac{1}{x})}$ or $\boxed{1+x^2}$

D. $\boxed{\frac{x+x^3}{\frac{1}{x}}}$ or $\boxed{x^2+x^4}$

E. $\boxed{-30\frac{1}{3}}$ or $\boxed{-\frac{91}{3}}$

F. $\boxed{129\frac{4}{5}}$ or $\boxed{\frac{649}{5}}$

G. $\boxed{5}$

H. $\boxed{2}$

⑧ A. $\boxed{2x^2-1}$

B. $\boxed{(2x-1)^2}$

C. $\boxed{7}$

D. $\boxed{49}$

⑨ A. $\boxed{\frac{1}{x} + (\frac{1}{x})^3}$ or $\boxed{\frac{1}{x} + \frac{1}{x^3}}$

B. $\boxed{\frac{1}{x+x^3}}$

C. $\boxed{\frac{17}{64}}$

D. $-\frac{1}{2}$

(10) i) 13

ii) $\frac{11}{3}$

iii) 4

iv) $\frac{10}{3}$

v) Undefined

vi) Undefined

(11) i) -12

ii) $\frac{1}{3}$

iii) Undefined

iv) -16

v) 3

vi) Undefined

(12) i) 10

ii) $\frac{3}{4}$

iii) 6

iv) 3

v) Undefined

vi) Undefined